KENYATTA UNIVERSITY

SPH 190 MECHANICS I

Dr. Margaret Chege

Physics Department

Kenyatta University

Course outline

- Units and dimensions. • Dimensional analysis
- Error analysis.
- Linear motion with constant acceleration; Displacement-time, velocity-time graphs, relative velocity.
- Projectiles
- Newton's Laws of motion.
- Friction.
- Static equilibrium

- Conservation of energy and momentum; Collisions.
- Circular motion
- Plane motion with constant acceleration
- Gravitation. .

1. Review

Units and dimensions

- Unit – a standard of reference/comparison for measurements.
- Fundamental quantities and SI units – cannot be derived from other units e.g. length (m), time (s) etc.
- Derived quantities and SI units – obtained by division of multiplication of basic quantities e.g. volume (m^3), density (kg/m^3) etc.
- Dimensions – used to determine if an equation is correct or determine units of unknown constants.

- Quantities may be assigned letters (dimensions) e.g. length L (or even m).
- Dimensions of all terms should be the same for equation to be considered correct.

Example:

1) Find the dimensions of the

following quantities a) Force

b) Work

c) Power

2) Using dimensional analysis determine the validity of the equation $x = \frac{1}{2} gt^2$ where x is the distance, g is the acceleration and t is the time..

Solution

1) Need to know the formula

c) $w = Fd$

Dimensions of $F = \frac{ML}{T^2}$

Dimensions of $d = L$

Dimensions of $W = \frac{ML}{T^2} L = \frac{ML^2}{T^2}$

b) $P = \frac{W}{t}$

Dimensions of $W = \frac{ML^2}{T^2}$

Dimensions of $t = T$

Dimensions of $W = \frac{ML^2}{T^2 T} = \frac{ML^2}{T^3}$

a) $F = ma$

$\underline{v}$ $d\,\underline{a} = t = t2$

Dimensions of $m = M$

L

Dimensions of a $=$ —

$TML2$

$\frac{1}{2}$ Dimensions

of F $=$ $T2$ 2) $x =$

gt^2

- Error analysis
- An error in a uncertainty in measurements.

Dimensions of $x = L$ Dimensions

of $g = \underline{\quad}^{L}$;

$T2$

Dimensions of $t^2 = T^2$

L

Dimensions of $gt^2 = \underline{\quad} T^2 = L$

$T2$

Dimensionally correct

- Indicates how far the measured (or calculated) value deviates from some true or value.
- Example, if repeat the same measurement several times, likely to get a slightly different value (larger or smaller) with every measurements.
- A value that may be considered as the true value of the measurement can be taken as the average of the measured values.
- Errors broadly categorised as random or systematic error.
- A random error is one that cannot be predicted or entirely avoided. e.g. change in room temperature affecting resistivity.
- Repeat measurements result in higher and lower values in equal measure
- Random errors can be quantified using statistical analysis.
- Systematic errors lead in values that are either high all through, or low all through relative to expected value. Example is using a poorly calibrated equipment.

Linear motion with constant acceleration

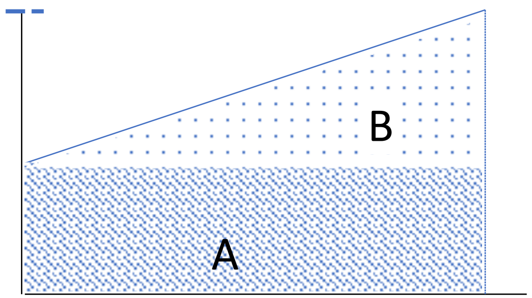
- Governed by three main equation v
- $v = u + at$ u
- $s = ut + \frac{1}{2} at^2$
- $v_2 = u_2 + 2as$ 0 t Time
- NOTE • $s = ut + \ v - u \ t$
- A body is said to be accelerating 2

  uniformly if the velocity changes • $s = ut + \ vt - \frac{1}{2} ut \ \frac{1}{2}$

  equally at equal intervals of time. •
- Area under the graph is equal to the $s = 2 \ ut + 2 \ \frac{1}{-} \ vt \ \frac{1}{-} = (-2 \ )t$ $u+v$

1 ( )

Velocity



distance covered

- From the diagram:
- $s = area\ A + area\ B$

- The term $\frac{u+v}{2} = v_{av}$ (average velocity).

- Hence
- $s = v_{av}t$

- Instantaneous velocity is the velocity at an instant in time
- For example instantaneous velocity at time $t = 0$ is u and at time $t = t$ is v. Examples

1) If the velocity (m/s) of a body travelling at a uniform velocity is:

$$v(t) = 2 + 3t$$

a) Find the instantaneous velocities at $t = 2\ and\ t = 3$

b) Find the average velocity between $t = 2\ and\ t = 3$

Solution

a) $v(t=2)=2+3(\times 2=)8\ m/s$

$v(t=3)=2+3(\times 3=)11\ m/s$

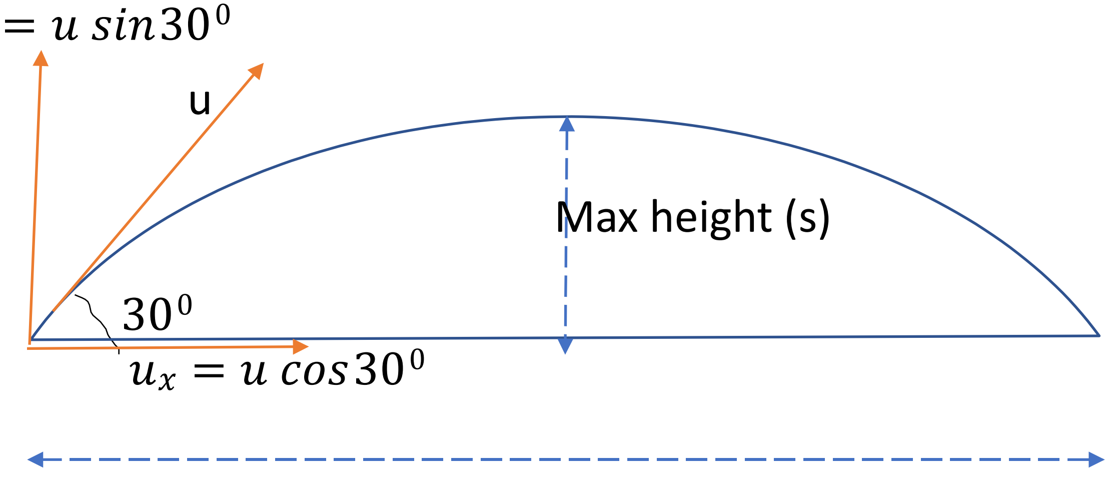
b) $v_{av}=\frac{8+11}{2}=9.5\ m/s$

2) A plastic projectile is to be launched at velocity of 50 m/s, 30^0 to the ground. Find (i) maximum height reached (ii) time of flight (iii) range Solution

-

A projectile is an object when launched follows a curved path.

The $= u\,sin30^0$



The launch velocity is resolved into two components, vertical component (u_y) and horizontal component (u_x)

•

$R = u_x t_f = u\,cos30^0 \times t_f$

u_y (i)

vertical component is responsible for the upward motion, falls to zero at max height. $v^2 = u^2 + 2as$

$y \quad y$

$v_y = 0\ and\ a = -g = -9.81\ m/s^2$

$0 = (50\sin 30^0)^2 - 2 \quad \times$

$9.81s$ s=31.86 m

(iii) $t_f = \frac{2u_y}{g} = \frac{2\times u\,sin\,30^0}{g} = \frac{2\times 50\,sin\,30^0}{9.81} = 5.1\ s$

(iv) The horizontal component is responsible for range, is constant (no horizontal accl.)

$R = 50\ cos30^0 \times 5.1 = 220.1\ m$

3) A ball is released from rest at height of 40 m above the ground . Find (i) velocity of the ball just before hitting the ground (ii) time it takes to hit the ground

Solution

i) $v^2 = u^2 + 2gs$

$v^2 = 2 \times 9.81 \times 40$

$\text{v} = \sqrt{784.8} = 28\, m/s$

ii) $s = ut + \frac{1}{2} gt^2$

$$t = \sqrt{\frac{2s}{g}} = \sqrt{\frac{2\times 40}{9.81}}\ 2.85\ s$$

Newton’s laws of motion

- The equations of linear motion describe the kinetics of an object, i.e. looking at motion of an object without considering the forces therein.
- Newtons laws of motion looks at the dynamics of an object, that is forces on an object as well as the consequences of that force e.g motion, equilibrium etc.
- There are three Newton’s laws of motion

i. First law: Law of inertia – a body remains at rest or in motion with constant velocity along a straight line unless acted upon by an external force.

ii. Second Law: The rate of change of momentum is directly proportional to the force applied and takes place in the direction of the force

- Momentum is product of mass and velocity. I
- If a body mass m moving with velocity u is acted upon by a force F so that in time Δt velocity changes to u, then;

- $F = \frac{mv - mu}{\Delta t}$ (i)
- $F = \frac{m(v-u)}{\Delta t} = m\frac{\Delta v}{\Delta t}$
- Now, $\frac{\Delta v}{\Delta t} = a(aceleration)$ hence

$F = ma$ (ii)

-

Note: equation (i) can be expresses as;

-

$$F\Delta t = mv - mu$$

(a) If Δt very small (e.g. bat striking a tennis ball) then;

$$F\Delta t = impulse \; Impulse = mv - mu$$

(b) If the net force is zero (F=0) then

$$0 = mv - mu \; mv = mu$$

Final momentum is equal to the initial momentum (or momentum is not changing with time) in which case momentum is said to be conserved.

Newton's 3rd law: action and reaction are equal and opposite

-

Examples

1. Two unequal masses m_1 = 15 kg and m_2 = 20 kg are connected by a string which passes over a frictionless pulley as shown. Find (i) the acceleration of the masses (ii) the tension in the string

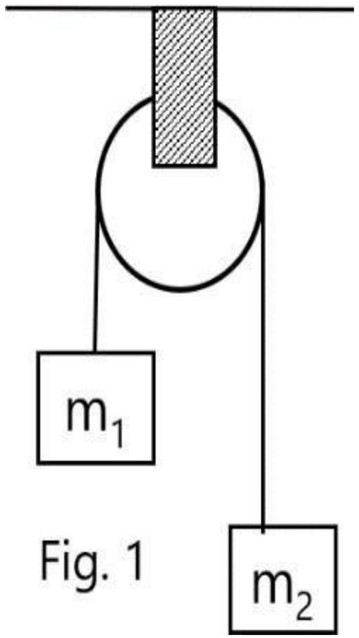


Fig. 1

- We redraw the diagram showing the forces (T is the tension
- Since the two masses cannot move in same direction, we let m_1 move up and m_2 down both with acceleration a. Hence:
- $T - m_1 g = m_1 a$ (i)
- $m_2 g - T = m_2 a$ (ii)

a) Adding equations (i) and (ii) gives:

$$(m_2 + m_1)a = (m_2 - m_1)g$$

$$a = \frac{(m_2 - m_1)g}{m_2 + m_1} = \frac{5 \times 9.81}{35} = 1.4\ m/s^2$$

b) $T - m_1 g = m_1 a$

$$T = m_1 g + m_1 a = 15 \times 9.81 + 15 \times 1.4 = 168.2N$$

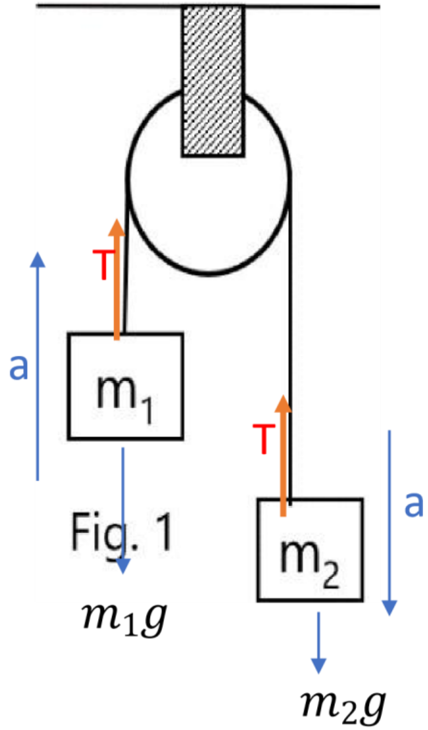


Fig. 1

2) A block of mass m_1 of mass 5 kg is placed on a smooth horizontal surface and pulled by a string attached to another block of mass m_2 of mass 5 kg hanging over a frictionless pulley as shown. Find (i) the acceleration of the (ii) the tension in the string

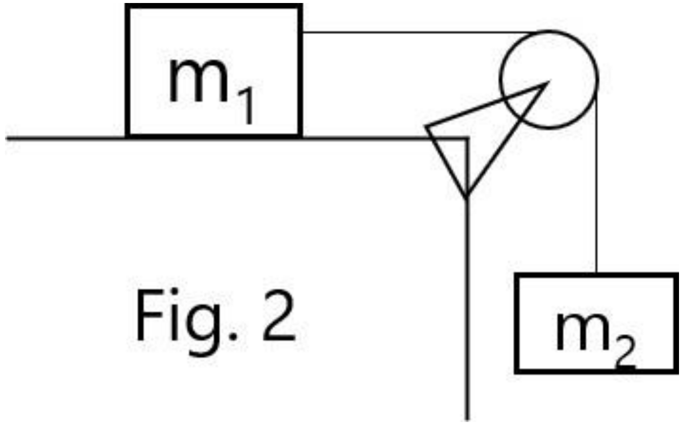


Fig. 2

$T = m_1 a$ (i)

$m_2 g - T = m_2 a$ (ii)

(i) Using (i) in (ii) and rearranging

$m_2 g = m_1 a + m_2 a$

$$a = \frac{m_2 g}{m_1 + m_2} = \frac{5 \times 9.81}{10}$$

$a = 4.905\ \mathrm{m}/s^2$

(ii) $T = m_1 a = 5 \times 4.905$

$T = 24.5\ N$

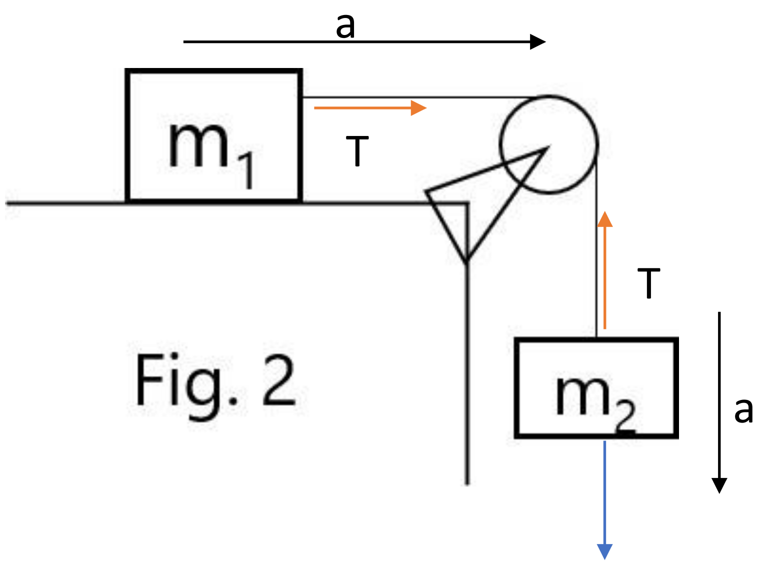


Fig. 2

- We redraw the diagram showing all the forces.
- Acceleration is in the $m_2 g$ direction of net force.
- Hence

2. ## Friction

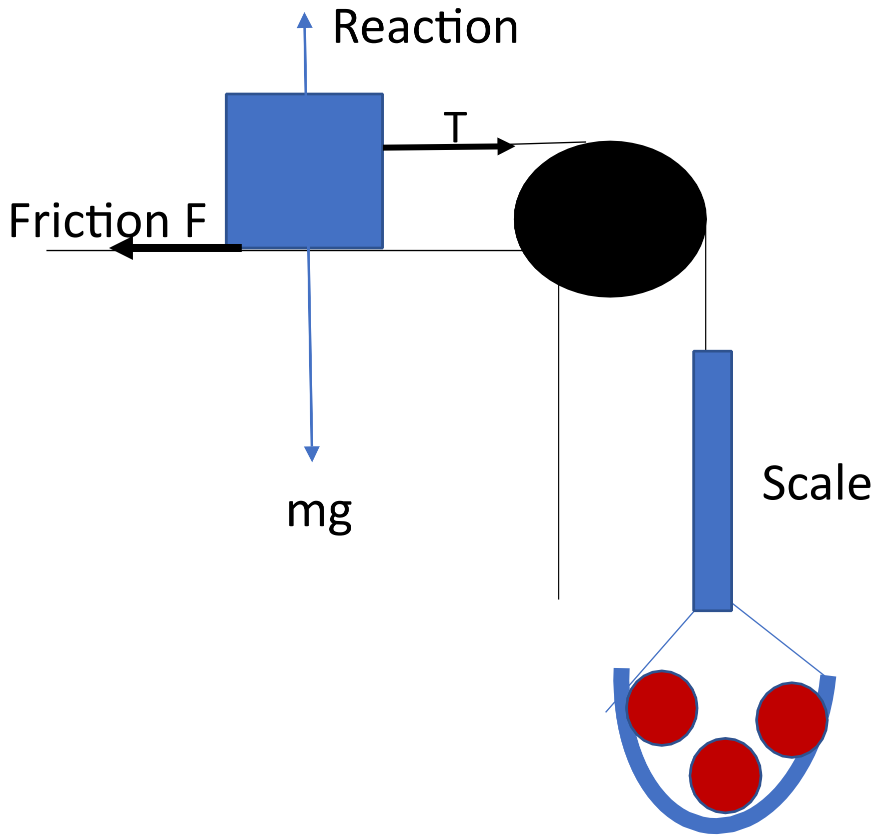
- Friction is opposition to motion between two surfaces that are attempting or attempting to slide over each other.

- Acts in direction opposite that of motion.
- Are two types;

i. Static/limiting friction

ii. Sliding/dynamic/kinetic friction Static friction

- Opposition force between two surfaces that are just about to slide over Scale each other. pan
- Consider the diagram shown which small weight added to the pan gradually.
- Initially the mass m does
- As more masses are added, a point is reached when the mass just starts to move.



- At this point the force applied (T) is equal to static frictional force (F).
- Static friction is proportional to the reaction (R=mg) and the nature of the surface.

Sliding friction

- This is the frictional force between two surfaces in contact and in relative motion.
- If the body is moving at a constant velocity, the force applied is equal to the sliding friction
- Sliding frictional force is usually lower than static force which means that more force is required to start a body moving compared to that required to keep the body moving.
- This is because some of the force applied is used to overcome inertia.

Laws of friction

- Experimental results on static and sliding friction are summarized into three laws:

i. The frictional force between two surfaces opposes their relative motion or attempted motion.

ii. Frictional forces are independent of the area of contact of the surfaces.

iii. For two surfaces that are not in relative motion, the static friction is directly proportional to the normal reaction. If the bodies are in relative motion, the sliding friction is directly proportional to the normal reaction and more or less independent of the relative velocity of the surfaces.

Coefficients of friction

- If F’ be static friction and F sliding friction of a body with normal reaction R, it follows from the third law of friction that:

$$F' = \mu' R \text{ and } F = \mu R$$

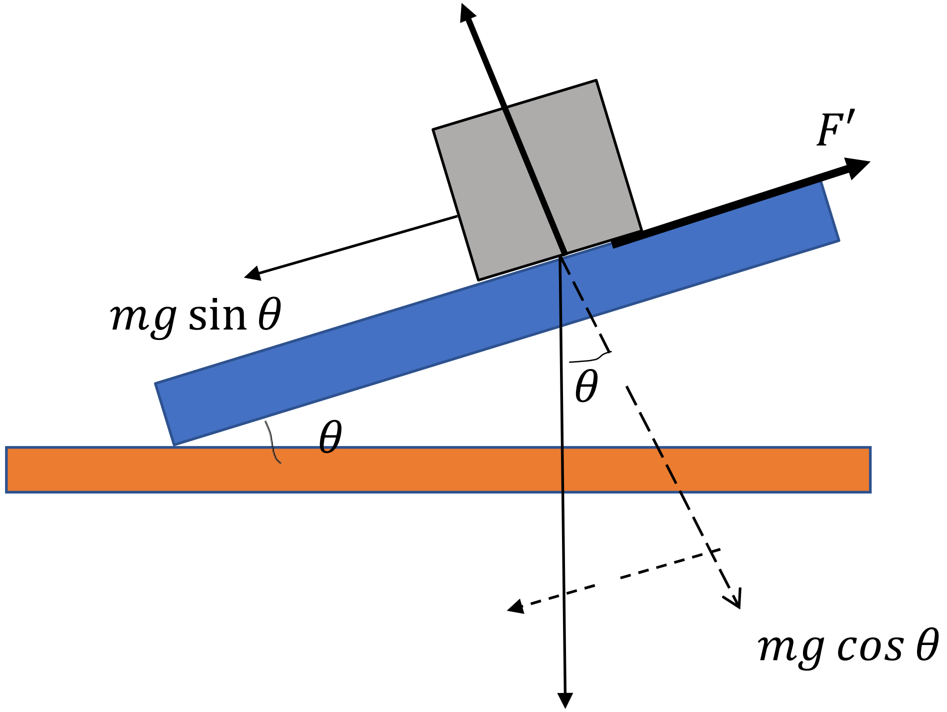
- μ' and μ are constants of proportionality referred to as coefficient of static friction and coefficient of sliding friction respectively.
- The coefficients depend on the nature and condition of the surfaces in contact.
- Example for steel on steel $\mu' \approx 0.8$ while for Teflon on Teflon, $\mu' \approx 0.04$.

- If two surfaces are assumed to be perfectly smooth, there is no frictional force and $\mu' = \mu = 0$.

Angle of friction and static friction R • Static friction can be determined

from

angle of friction from the arrangement shown.



- The tilted plank is raised gradually and the angle θ at which the mass starts to slide determined.
- This angle is called angle of friction.

- At this point, the coefficient of friction is;

$$\mu' = F\_R' = \_\_\_\_\_\_mgmg \text{ cossin } \theta\theta \qquad mg$$

$$\mu' = \tan \theta$$

Example 1

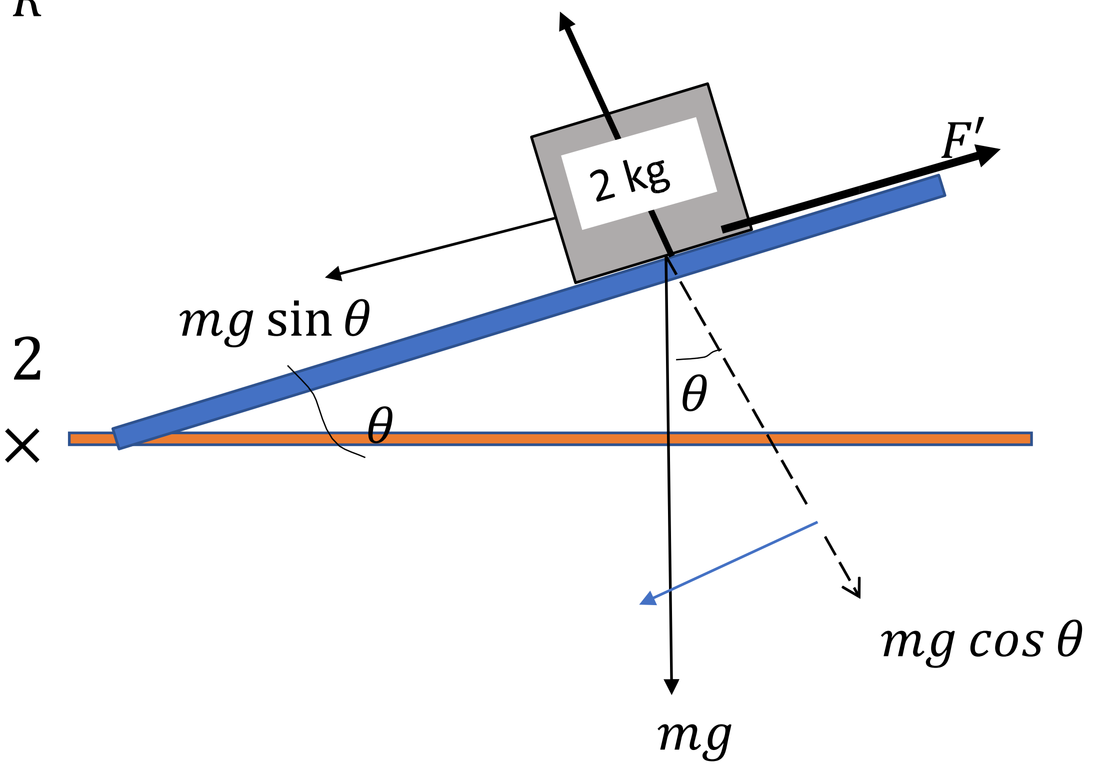
A block of mass 2 kg rests on a plane lying on a horizontal surface. When one end of the plane is lifted gradually, it is found that the block is on the verge of slipping when the plane is inclined at an angle 35^0. Determine the

i. value of coefficient of static friction between the block and the inclined plane.

ii. The frictional force iii. The reaction

Solution

i. $\mu' = \tan\theta = \tan 35^0 = 0.7$ ii. $F' = mg\sin\theta = 2 \times 9.81 \sin 35^0 = 11.25\ N$ iii. $R = mg\cos\theta = 2 \times 9.81\cos 35^0 = 16.07\ N$



Example 2

A 1500 kg car moves along a straight horizontal road with an initial speed 140 m/s when it brakes sharply then slides. If the coefficient of kinetic friction 0.8, calculate: i. the deceleration ii. Time it takes the car to come to rest iii. the distance travelled by the car before it come to rest iv. What would the distance be if the car was moving at 80 m/s?

$\Longrightarrow a = \underline{\qquad} = \mu g$

$m \quad {}_2\, a = 0.8 \times 9.81 = -7.85\ m/s$ ii) $v = u$

$+ at$

Solution ii) Friction slowing down car

$F_{friction} = \mu R = -ma$

μmg

$0t \ == \ 17.84140 - s\ 7$

$2 + \times 2\ 7.85as\ s$

iii) $v^2 = u^2 + 2as$

$0 = 140^2 - 2 \times 7.85s$

$s = 1248.41$ m

.85t iv)$0 = v\ {}_2 80 =_2\ u -_2$

$s = 407.64\ m$

Static equilibrium

Solution

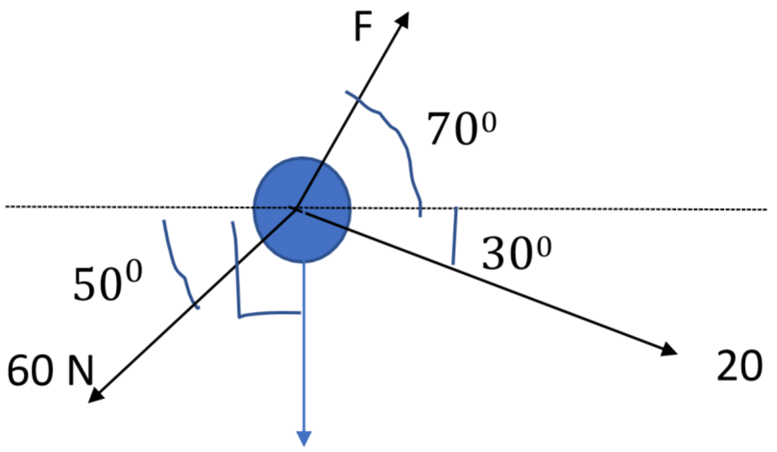
- This means that all forces that are acting on Reduce the forces into vertical and horizontal components.

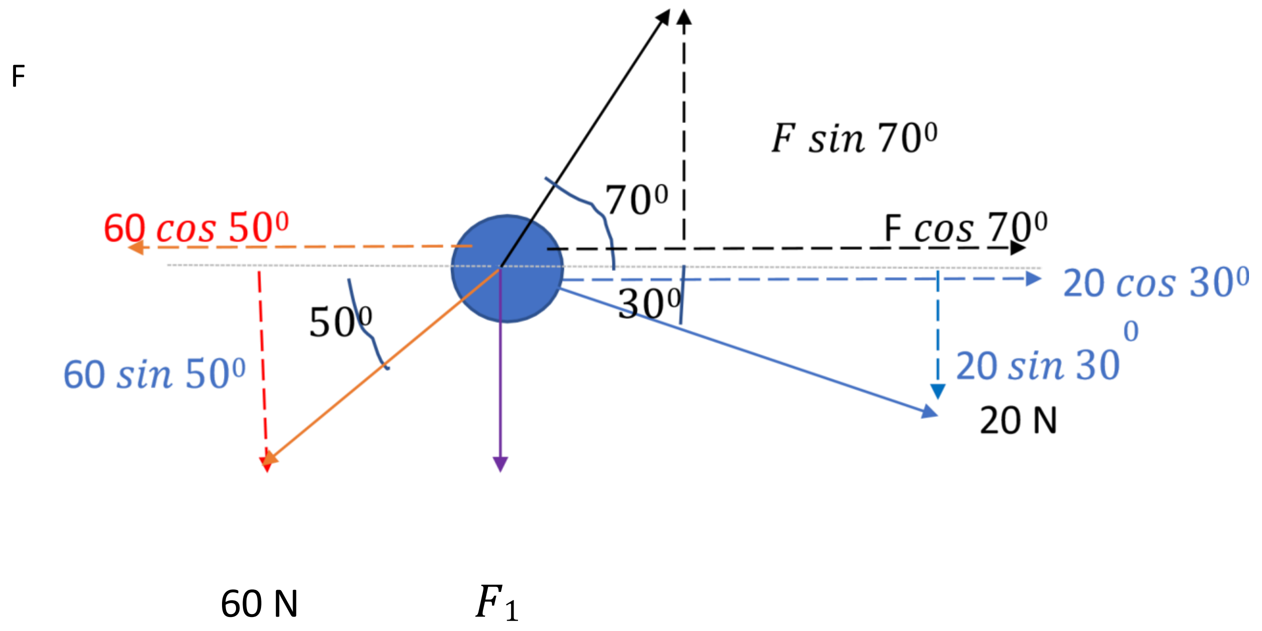
an object balanced and therefore the object is not moving. Magnitudes of left horizontal components rightcomponents; Same as vertical up and down ones.

- The resultant force on an object is zero.

Example 1

An object is acted upon by three forces as shown.
(a)If the system is in static equilibrium,
find the value of force F $and\ F_1$.





(a) $60\cos 50^0 = 20\cos 30^0 + F\cos 70^0$

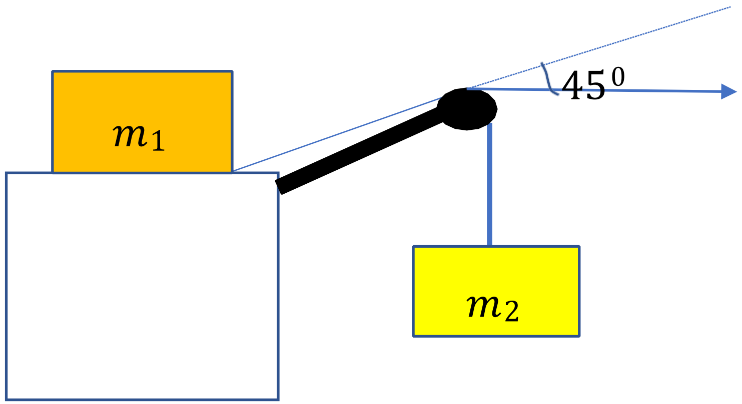
$F = 62.12\ N$

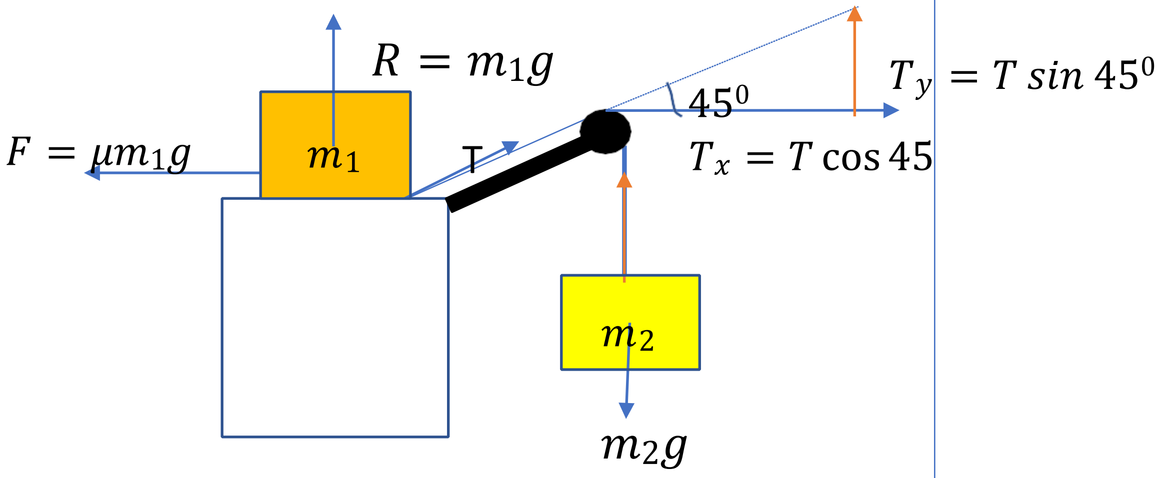
N (b) Vertical $60 \sin 50$ components$_0$ $+ 20 \sin\ 30$will$_0$ return$+ F_1 =$ the 62.12 samesin value 70_0

F_1 $F_1 = 2.41\ N$

Example 2

- A block m_1 of mass 27 kg is resting on a horizontal table as shown. The coefficient of static friction between the block and the table is 0.45. Determine the maximum mass a second block for which the system should be at equilibrium





resolve then to horizontal and vertical

components. At static equilibrium:

Resultant horizontal is zero hence:

$\mu m_1 g = T \cos 45^0$ (i)

Rersultant vertical forces is zero hence:

$m_2 g = T\, sin\, 45^0$ (ii)

Conservation of linear momentum

Divide (ii) by (i)

$$\frac{m_2}{\mu m_1} = \tan 45^0$$

$$m_2 = \mu m_1 \tan 45^0 = 0.45 \times 27 \tan 45^0$$

$$m_2 = 12.15\ kg$$

We first draw all the forces in the system,

- Suppose two bodies A and B moving towards each other at constant velocity are involved in a collision (say no external force exists)
- According to Newton’s third law of motion, force on A due to B equals force on B due to A.
- This means that each body experiences the same rate of change of momentum but in opposite directions hence the total change in momentum is zero.
- The momentum is then said to be conserved in accordance with the law/principle of conservation of linear momentum.
- This states that the total linear momentum of a system of interacting particles/bodies in which no external forces are acting remains constant.

Examples:

A body A of mas 4 kg moves with a velocity of 2 m/s and collides head on with another body of mass 3 kg moving in opposite direction at 5 m/s. After collision, the bodies move off together with a velocity v. Find v.

Solution

Momentum before collision = momentum after collision

Take direction of A to be positive and that of B negative Assume after collision both move in the positive direction.

Then

$(4 \times 2) - (3 \times 5) = (4 + 3)v$

$8 - 15 = 7v$

$v = -1\ m/s$

This means that the bodies move in the direction B was moving before collision.

Example 2

A bullet of mass m_b is fired horizontally from a gun of mass m_g. Find the recoil velocity of the gun if the velocity of the bullet is v_b.

Solution

Before firing, both gun and bullet are station

After firing, bullet moves away from the gun (+ve direction) with velocity v_b.

- To conserve momentum, gun must move in the opposite direction (-ve direction) with velocity $-v_g$.
- Hence
- $Momentum\ before = momentum\ after$
- $0 = m_b v_b - m_g v_g$
- $v \quad = \frac{mb}{mg} v_b$

Example 3

- A flat truck of mass 400 kg is moving freely along a horizontal track at 3 m/s. A man moving at right angles jumps onto the truck causing its speed to decrease by 0.5 m/s. What was the mass of the man Solution

- since man was moving perpendicular to the motion of truck, he had zero momentum relative to the truck –hence initial momentum only that of truck.
- After he jumped on the truck, both man and track moved at same velocity .
- If mass of man m, then;
- $Momentum\ before = momentum\ after$
- $0 + 400 \times 3 = (400 + m) \times 2.5$
- $1200 = 1000 \times 2.5m$
- $m = \frac{1200-1000}{2.5} = 80\ kg$

Elastic and in-elastic collision

- When two bodies collide in the absence of external forces momentum is conserved.
- However the impact converts some of the kinetic energy of the colliding objects to heat, sound, deformation work etc which means that the combined kinetic energy is usually less that the final combined kinetic energy.

- Such a collision where momentum is conserved but kinetic energy is not is called in-elastic collision.
- Some collisions are such that the kinetic energy lost is so small that it can be considered insignificant e.g. tennis ball hitting a racket.
- Such collisions where both kinetic energy and momentum are conserved are called elastic collisions.
- Note: the ratio of the relative velocity of separation $(v_2 - v_1)$ to the relative velocity of approach $(u_1 - u_2)$ coefficient of restitution (e).
- $e = \frac{v2 - v1}{u1 - u2}$
- If $v_2 = v_1$ then $e = 0$ and he collision is perfectly in-elastic
- If $v_2 - v_1 = u_1 - u_2$ the $e = 1$ and the collision is perfectly elastic.

Examples

A body A of mass 6 kg moving at a velocity of 9 m/s collides with a body B mass 3 kg moving in the same direction as A with a velocity 4 m/s. If the velocities of A and B after colliding are $v_A\, and\ v_B$ respectively and the coefficient of restitution 0.8, find $v_A\, and\ v_B$. Assume no external forces act on the system.

Solution

Momentum is conserved hence

$Momentum\ before = momentum\ after$

$(6 \times 9) + (3 \times 4) = 6v_A + 3v_B$

$66 = 6v_A + 3v_B$ (i)

Now $4 = v_B - v_A$

$$e = \frac{v_2 - v_1}{u_1 - u_2}$$

$$0.8 = \frac{v_B - v_A}{9 - 4}$$

$4 = v_B - v_A$

$v_B = 4 + v_A$ (ii)

Using (i) n (i)

$66 = 6v_A + 3(4 + v_A)$

$54 = 9v_A$

$v_A = 6\ m/s$

$4 = v_B - 6$

$v_B = 10\ m/s$

Example 2 Conservation of KE

A tennis ball moving at 10 m/s along x-axis 1

collideswhich is elasticallyat rest. Assuming with an

one ball- $1002\, m \times = 10 v_2 \quad += v\, 2_{22}\ m$(ii()$v_1 +$

dimensional, determine the final velocity of each 1

of the two balls. Now, from eq. (i)

Solution $v_1 = 10 - v_2$

Let each ball have mass m $v_2 = 100 - 20 v_2 + v_2$ (iii)

$1 \qquad 2$

Let the of the balls after collision (assume moving

in +ve x direction be $v_1\, and\, v_2$ respectively $100 =$

$2 \qquad 2 \quad 2$

$- \qquad 2 \quad - \quad 1 \quad 2 \quad 2$

identicalthe collision tennis is

$v_2\,)$

$$v_1^2 = (10 - v_2)^2$$

Use (iii) in (ii)

$100 - 20v \qquad + v_2 + v_2$

Both momentum and kinetic energy are $20v_2 = 2v_2^2$ conserved hence;

$v_2 = 10$ m/s

Conservation of momentum $v_1 = 10 - v_2$

$m \times 10 = m(v_1 + v_2)$ $v_1 = 10 - 10 = 0$

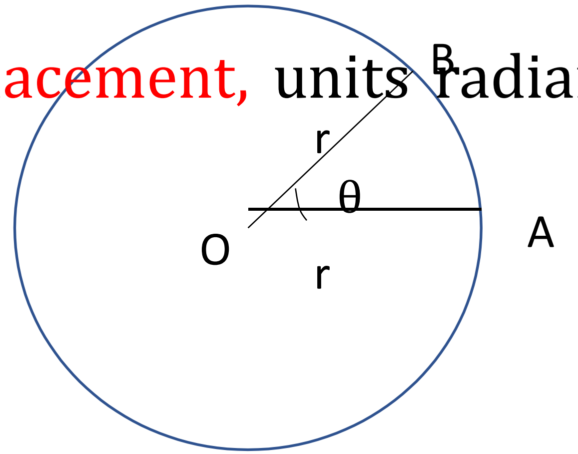
$10 = v_1 + v_2$ (i)

Circular motion

Definition of terms

- Say an object moves in a circular orbit and after some time t, the particle long the arclength AB sweeping an angle θ about O.
- The angle swept by the particle is called angular displacement, units radians (rad)
- The rate of change of angular displacement with time is the angular velocity, ω, units rad/s. Thus;

$$\omega = \frac{\theta}{\qquad} \quad (i)$$



t

- The time taken by the object to complete 1 revolution is called period, T
- Since the angular displacement of the object after 1 revolution is equal to 2π , then;

$\omega = \_\_^{2\pi}\ T$ (ii) $T =$

$\_\_^{2\pi}$ (iii)

ω

- Frequency, f (Hz) is the number of complete revolutions in unit time and given by: 1

$f = \_$ (iv)

T

- Using this in eq. (ii) give;

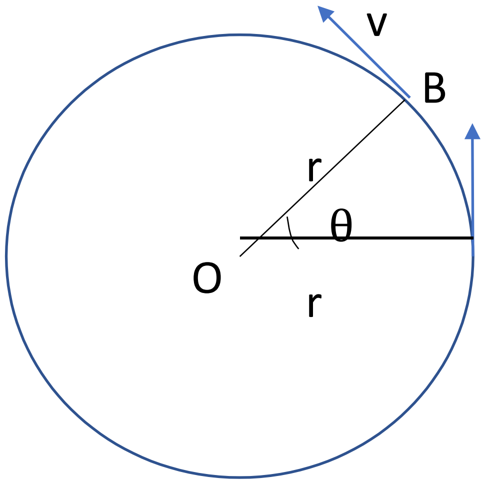
$\omega = 2\pi f$ (v)

- Linear velocity v (m/s) is the velocity the object would move at if for some reason it abruptly stopped moving in the circular path e.g. strong whirling a stone breaks.
- The linear velocity (referred to as just velocity) is tangential to the path of rotation.
- If the object is moves along the arclength AB with velocity v, then:

$$v = \frac{arclength\ AB}{t} \quad \text{(vi)}$$

v

A



- Now $arclength\ AB = \theta r$. Using this in (iv) gives

$$v = r\frac{\theta}{t}$$

$$v = r\omega \qquad \text{(vii)}$$

Example

A particle in uniform circular motion makes 2 revolutions in 0.5 s seconds. If the circular path has a radius of 2 m, find: (a) period (ii)frequency (iii) angular velocity (iv) linear velocity. Solution

(i) $T = \frac{time}{tcomplete\ revolutionse} = \frac{0.5}{2} = 0.25\ s$

(ii) $f = \frac{1}{0.25} = \frac{1}{0.25} = 4\ Hz$

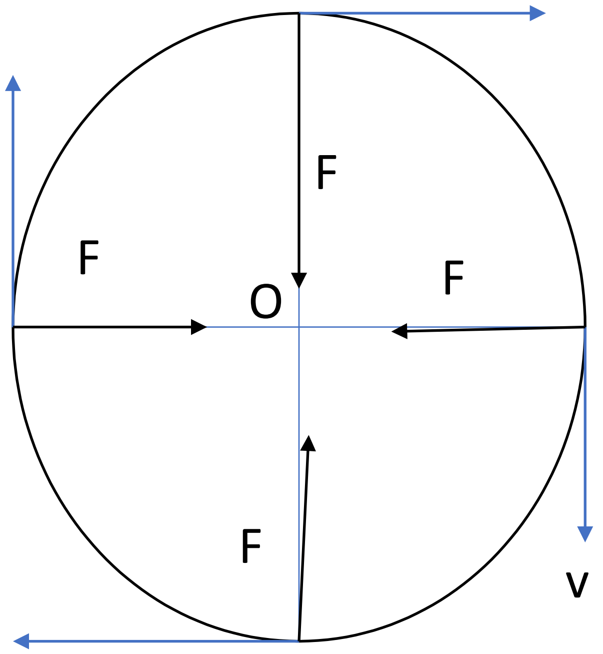
(iii) $\omega = 2\pi f = 8\pi\ rad/s$

(iv) $v = r\omega = 16\pi\ m/s$ **Centripetal force**

- If a body is moving in a circular path, there must be a force keeping it in the path otherwise it would fly away in a straight line with constant velocity in accordance to newtons second law of motion.

- If the body is moving at a constant speed (v), the force cannot be in the direction of motion since it would end up increasing the speed.
- The force must therefore be perpendicular to the direction of motion of the body hence must be direction towards the center of rotation.

- This force is called centripetal force (F). v v
- For a stone tied to a string being whirled in a circular path, the centripetal force is provided by tension in string.
- If the string breaks, stone flies away.
- For planets orbiting the sun, the centripetal force is the gravity.



- For electros orbiting electrons, the centripetal force is electrostatic. v

Centripetal acceleration

- Since there is a resultant force acting on a body in circular motion, then by Newtons second law the body must have an acceleration.
- This acceleration, called centripetal acceleration, is directed toward the axis of the circular path just like the centripetal force.
- For a body moving with constant angular velocity ω in a path of radius r, the centripetal acceleration is given by;

$$a = \omega^2 r \qquad \text{(viii)}$$

- Since $v = r\omega \Longrightarrow \omega = \frac{v}{r}$ where v is the linear velocity equation (viii) may be written as;

$$a = \frac{v2}{r} \qquad \text{(ix)}$$

Proof

- Consider a particle moving at a constant speed v in a circular path radius r.
- Consider the particle motion at PQR.
- The x-component of the velocity of the particle has the same value at P and R.
- The x-component of acceleration is therefore zero.

- The y-component of velocity at P is $-v \sin\theta$ (-ve since facing down) while at R it

is $v \sin\theta$ (+ve since facing up).

- The y-change in velocity from P to R is thus;

$$v \sin\theta - (-v \sin\theta) = 2v \sin\theta$$

- If it takes time t to move from P to R, then;

$a = \frac{2v \sin\theta}{t}$ (i)

- The speed (v)of the particle along the arc PQR:

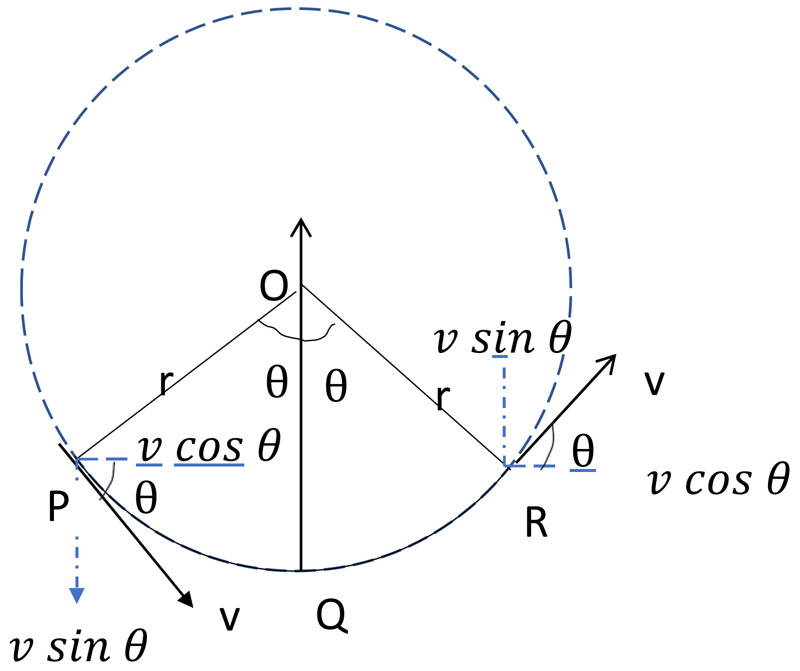
$v = \frac{arclength\ PQR}{t}$

$= \frac{2\theta r}{t}$ (ii)

$t = \frac{2\theta r}{v}$ (iii)

- Using (iii) in (i) leads to;

$$a = \frac{v^2}{r}\frac{\sin\theta}{\theta} \qquad \text{(iv)}$$



- If θ is very small, $\sin\theta \approx \theta$, eq. (iv) becomes;

$v2$

$a = \_$ (v) r

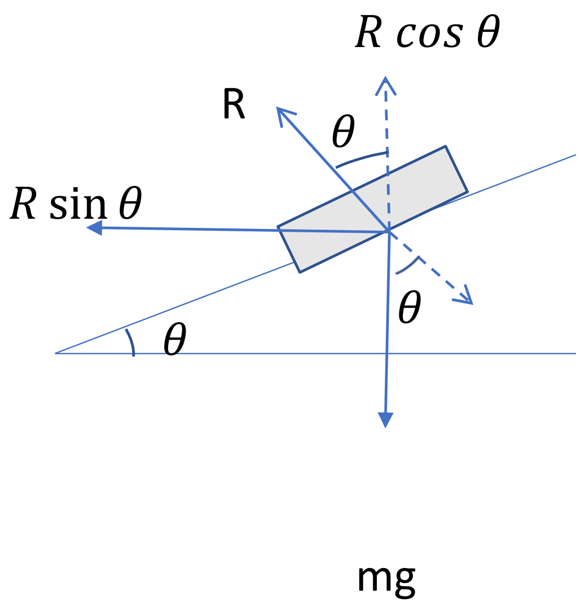
Vehicles going round bends

- Centripetal force required by a car to go round a bend on a level ground is provided by the frictional force between the car tyres and the road.
- Need to rely on friction is however removed if the road is suitably banked (tilted through angle θ, called banking angle).
- The normal reaction R acquires a horizontal component $R \sin \theta$ due to banking.
- If m be the mass of the car; r radius of circular path, and v the speed centripetal force to provide acceleration become is equal to the horizontal component of R i.e.

$$R \sin \theta = \frac{mv^2}{r} \qquad \text{(i)}$$

- From the diagram;

$$R \cos \theta = mg \qquad \text{(ii)}$$



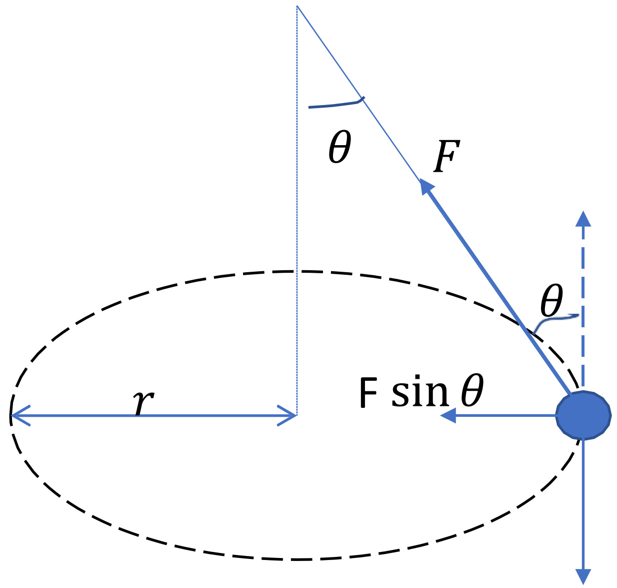
- Dividing eq. i) by (ii) leads to:

$$\tan \theta = \frac{v2}{rg} \qquad \text{(iii)}$$

The conical pendulum

- A pendulum is displaced sideways then given a small push at right angle to its displacement so that it moves in a horizontal circle, strong sweeps out a cone.
- The pendulum (mass m) bob is acted upon by two forces (i) weight mg and (ii) tension.

- Resolving the tension F gives the horizontal component $\mathrm{F}\sin\theta$ directed towards the center of rotation (radius r) and a vertical component $\mathrm{F}\cos\theta$.
- The horizontal component is equal to the centripetal force and if v the velocity, then; hence;
- $\mathrm{F}\sin\theta = \frac{mv^2}{r}$ (i) $F\cos\theta$
- There is no versicle acceleration hence;
- $F\cos\theta = mg$ (ii)
- Dividing (i) and (ii) gives;
- $\tan\theta = \frac{v2}{rg}$ (iii)
- This is same as previous equation mg



Period of a conical pendulum

- Now, since $v = r\omega$ equation (iii) may be written as;

$$\tan\theta = \frac{r2\omega 2}{rg} \qquad \text{(iv)}$$

2π

$$\tan\theta = \frac{r\omega^2}{g} \qquad \text{(v)}$$

- Now, if L be the length of the string

$r = L\sin\theta$. Also $\tan\theta = \frac{\sin\theta}{\quad}$.

- But $\omega = \frac{2\pi}{T}$ hence;

$$\frac{2\pi}{T} = \text{(viii)} \sqrt{\frac{g}{l\cos\theta}}$$

$$\sqrt{\frac{L\cos\theta}{g}}$$

$\cos\theta$

- Using these in (v) $\sin\theta$

$\omega^2 L \sin\theta$

$$T = 2\pi \qquad \text{(ix)}$$

$=$ (vi)

$\cos\ \theta$ g

$$\omega = \sqrt{\frac{g}{l\cos\theta}} \qquad \text{(vii)}$$

Example

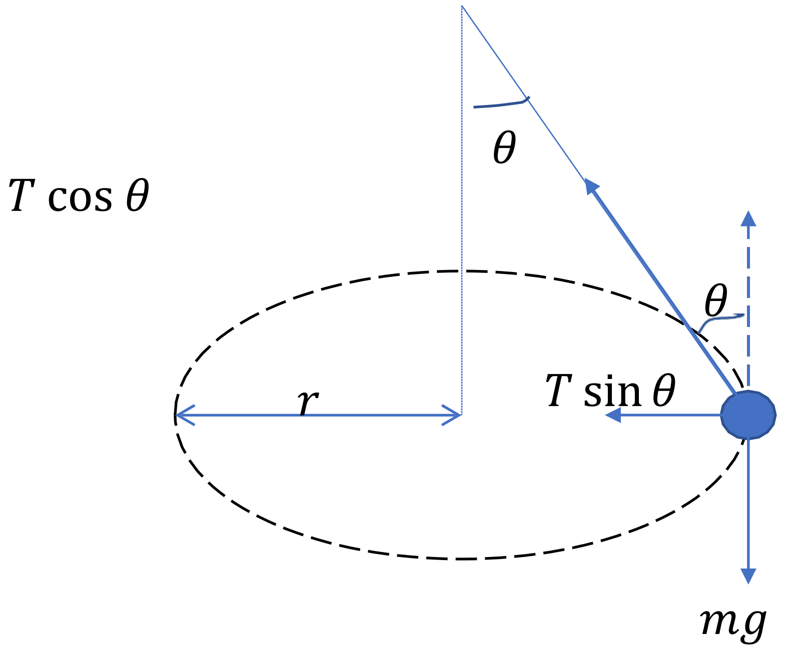
A pendulum bob of mass 2 kg is attached to one end of a string of length 1.2 m. The bob moves in a horizontal circle in such a way that the string is inclined at 30^0 to the vertical. Calculate (a) the tension in the string (b) the period of motion. Take $g = 9.81 \frac{m}{s^2}$.

Solution

(a) $F \cos 30^0 = mg$

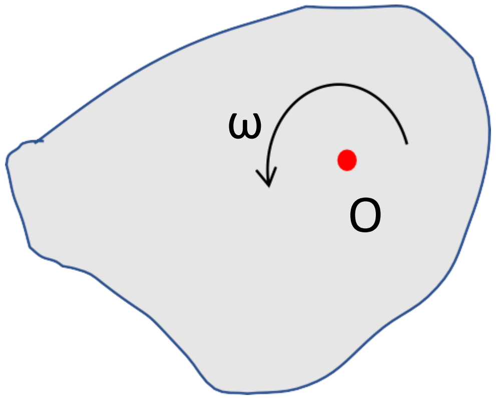
$$F = \frac{mg}{\cos 30_0} = \frac{2 \times 9.81}{0.866} = 22.6\ N$$

(b) $$T = 2\pi\sqrt{\frac{L\cos\theta}{g}} = 2\pi\sqrt{\frac{1.2\times\cos 30^0}{9.81}} = 2.04\ s$$



Plane motion with constant acceleration

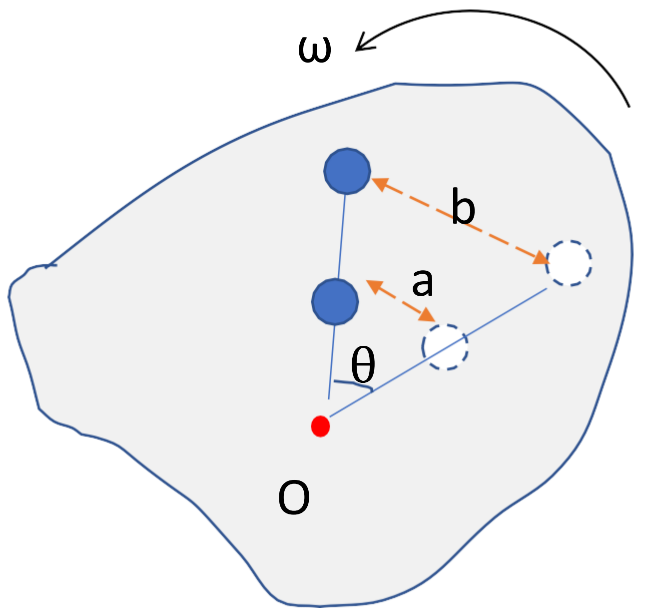
- When a body/plane rotates about a fixed point (O), it posses kinetic energy due to its rotation.
- If ω be the angular velocity, then the rotational kinetic energy is given by;
- $KE_{ro} = \frac{1}{2} I\omega^2$ (i)
- Where I ($kg\ m^2$) is a constant for a given axis of rotation called moment of inertia of the rotating body.



Proof of rotational kinetic energy equation

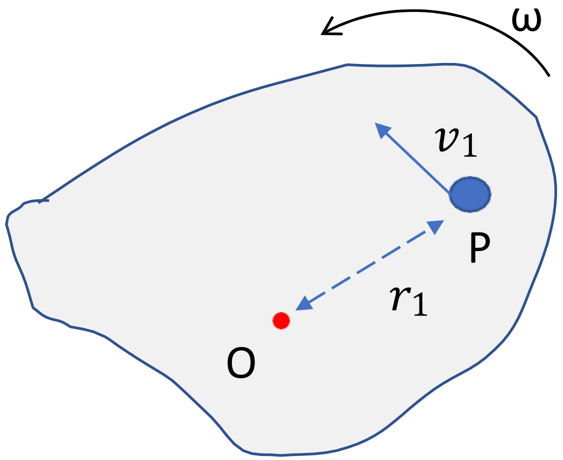
- Consider a body rotating at a constant angular velocity ω about some axis O.
- Suppose that the body is made of n particles each of mass m_i, $i = 1,2,3 \ldots n$.
- All the particles have the same angular velocity as they move through equal angular displacements in equal times.

- However, particles closer to axis of rotation cover shorter linear distances compared to particles further away within the same time span
- This means that the linear velocities of the particles is not uniform.
- Particles closer the axis of rotation travel at lower speeds compared to particles further away.



- In a see-saw for instances, kids two sit further away from the fulcrum swing faster than those close to the fulcrum.

- Now, consider a particle at P, mass m_1 and distance r_1 from the axis of rotation.
- If the particle has a linear velocity v_1 then;
- $v_1 = r_1\omega$
- The kinetic energy of the particle is thus;
- $$KE = \frac{1}{2}m_1v_1^2 = \frac{1}{2}m_1r_1^2\omega^2$$



- If the rest of the particles have masses $m_2, m_3 \ldots \ldots m_n$ and there distances from the axis of rotation $r_2, r_3 \ldots \ldots r_n$ respectively, then the total kinetic energy due to rotation is.
- $$KE_{rot} = \frac{1}{2}m_1r_1^2\omega^2 + \frac{1}{2}m_2r_2^2\omega^2 + \cdots + \frac{1}{2}m_nr_n^2\omega^2$$

- $KE_{rot} = \frac{1}{2}\omega^2(m_1r_1^2 + m_2r_2^2 + \cdots + m_nr_n^2)$
- For a given rotating body, the term in the bracket is constant and is called moment of inertia, I.

$_1r^2_1 + m_2 \ {}^2_2 + \cdots +$ • $I = mr \ m \ r^2 = \sigma^n \ m_ir^2$

$n \ n \qquad i=1 \qquad i$

•• Hence;KE $1 \quad 2$

$rot = \_2 \ I\omega$ Note:

- The moment of inertia is a measure of the way mass is distributed about ti center of rotation.
- It depends on the shape of the object (plane, spherical, cylindrical etc.), the mass of the rotating object and the axis of rotation (e.g. about an axis passing perpendicularly through the body, about one end etc

Angular momentum

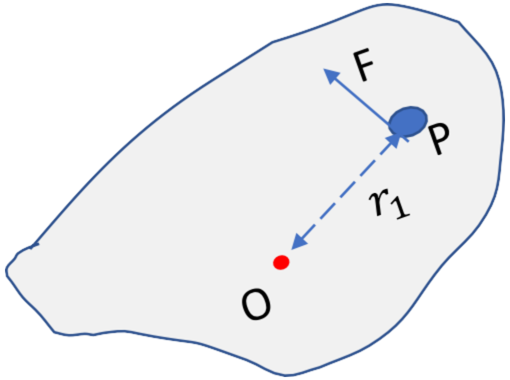
- For a rigid body rotating about an axis, all particles have the same angular velocity ω but linear velocity v depending on the distance r from the axis of rotation.
- Each particle has an associated linear momentum P given by
- $P = m_i v_i = m_i r_i \omega$

  where i=1,2,3….
- The angular momentum L of a particle about the axis of rotation is defined as the product of its linear momentum and the perpendicular distance between the particle and the axis of rotation, that is;
- $L_i = P_i r_i = m_i r_i^2 \omega$
- For n particles, total angular momentum given by:
- $L = \sum_{i=1}^{n} P_i r_i = \omega \sum_{i=1}^{n} m_i r_i^2$
- $L = I\omega$

- For a rotating body, angular momentum is conserved if there is no external force acting on it.

Torque and Newton's second law

- If a rigid body is acted by a force F perpendicular to the distance r from the axis of rotation, the force makes the object move in a rotational motion.
- The product of the force and the distance to the axis of rotation is referred to as torque τ;
- $r = rF$
- When torque is applied, the rotating body accelerates hence linear momentum changes with time.



- By Newtons second law, torque is change of angular momentum with time hence;
- $$r = \frac{dL}{dt} = \frac{d}{dt}(I\omega)$$

- If the moment of inertia is constant, then;
- $$r = I\frac{d\omega}{dt}$$
- $$r = I\alpha$$
- Where α is the angular acceleration

Comparison of linear and rotational/plane quantities

	Linear motion	Plane motion
Amount	Mass, m	Moment of inertia, I
Velocity	v	
displacement	s	

Acceleration	a	ω
		θ
		α
KE	$KE = \frac{1}{2} mv^2$	$KE_{rot} = \frac{1}{2} I\omega^2$
Momentum	$P = mv$	$L = rP\ (P \perp r);\ L = I\omega$
Force	F	$r = Fr\ (F \perp r)$
Newton’s second law	$F = ma$	$r = I\alpha$
Work	Fr $(F \parallel r)$	$W = r\theta;\ W = Fr\theta$

equations of linear and rotational motion

$v = u + at$	$\omega_f = \omega_i + \alpha t$
$s = ut + \frac{1}{2} at^2$	$\theta = \omega_i t + \frac{1}{2} \alpha t^2$
$v^2 = u^2 + 2as$	$\omega_f^2 = \omega_i^2 + 2\alpha\theta$
$s = \frac{u + v}{2} t$	$\theta = \frac{\omega_i + \omega_f}{2} t$

Examples

A flywheel is mounted on a horizontal axle with a radius of 0.06 m. A constant force of

50 N is applied tangentially to the axle. If the moment of inertia o the system (flywheel and axle) is 4 $kg\ m^2$, calculate (a) the angular acceleration of the flywheel
(b) the number of revolutions the flywheel makes in 16 s assuming it starts from rest.

Solution

(a) we use the two torque formula

$$r = rF = 0.06 \times 50 = 3\ Nm$$

$$r = I\alpha$$

$$\Rightarrow \alpha = \frac{c}{I} = \frac{3}{4} = 0.75\ rad/s^2$$

(b) Use equations of motion

$$\theta = \omega_0 t + \frac{1}{\ } \alpha t^2$$

$$\theta = 0 + 0.75 \times \frac{1}{2}^{\,2} \times 16 = 96\, rad \qquad {}^{2}$$

$$1\ revolution = 2\pi\ rad$$

$$Number\ of\ revolutions = \frac{96}{2\pi} \approx 15$$

Gravitation

Kepler's laws

- These are the three laws that describe planetary motion:

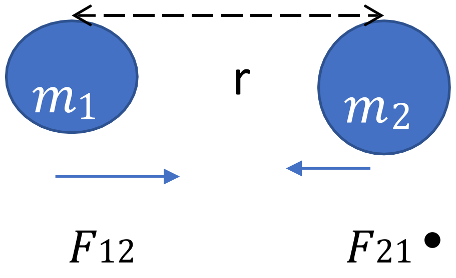
1) The orbit of each planet is an ellipse which has the sun as one of its foci
2) Each planet moves in such a way that the imaginary line joining it to the sun sweeps out equal areas in equal time
3) The squares of period of revolution of the planets about the sun are proportional to the cubes of their mean distances from it.

Newton's law of Universal gravitation

- Years after Kepler's laws were announced, Isaac Newton showed that any body that rotates around the sun must be acted upon by a force directed towards the sun.
- He proposed that this force is similar to that which keeps the moon in its orbit around the earth, and that which the earth exerts on a body that causes it fall to the ground.
- He also proposed that every object in the universe attract another object with a force that is inversely proportional to the square of the distance between them.
- Since each body exerts equal force on the other, Newton proposed that the force between the bodies must be proportional to the product of the masses.
- These propositions are summarised in Newton's law of universal gravitation which states that:

- Every particle in the universe attracts another with a force that is proportional to the product of their masses and inversely proportional to the square of the distance between them.

- $F \propto \frac{m_1 m_2}{r^2}$ ____(viii)



- $F = G\frac{m_1 m_2}{r^2}$ (ix)
- G is a constant of proportionality called universal gravitational constant.
- $F_{12} = -F_{21}$
- Say force is of magnitude F. Then;
- $G = 6.67 \times 10^{-11}\ Nm^2/kg^2$

Example

Show that Kepler's third law is consistent with newton's gravitational law $F = G \frac{m_1 m_2}{r^2}$

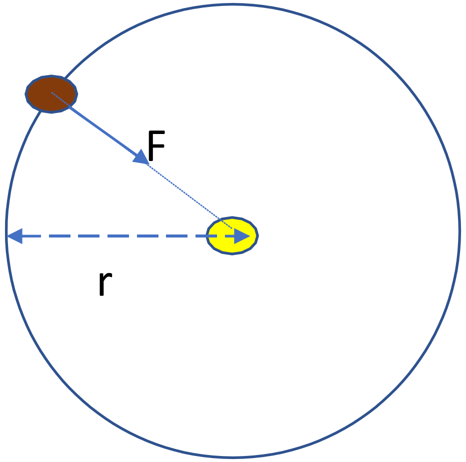
Solution

- Consider a planet mass m_p moving about the sun, mass m_s in a circular orbit radius

$$\frac{m \;\; m_2}{r^2}$$

- $F = m_e\omega^2 r$
  - This force is equal to the gravitational force between the sun and earth;

r with angular velocity ω.



- The centripetal force of he earth as it

- $F = G\_m^{ems}$

  $r2$

- Hence
  - $m\ \omega 2r = G\_me2m\ s$

  $^2 =$ 3 $G\ m^s\ r$

  - ω 2 π r

  - Now ω $= T$

  hence $^2 = 4\pi^2$

  $r3 \bullet T$

  $Gm^2{}_s$

4π

- Since_ = $constant$

then;

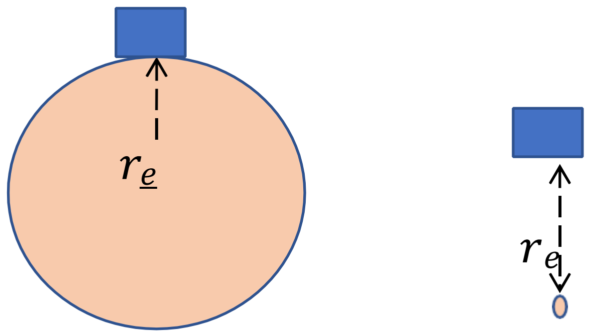
Gms

- $T^2 \infty r^3$

moves round the sun is given by; • This is Kepler's third law.

Mass of the earth

- If a body of mass m is on the surface



of the earth, the force on the body due to force of gravity is;

- $F = mg$
- The force is equal to that given by the universal law of gravitation;
- $F = G \frac{m_e m}{r_e^2}$
- Hence
- $G \frac{m_e}{r_e^2} = g$
- $m_e = \frac{r_e^2 g}{G}$

Example

Using newton's law of universal gravitation, estimate the mass of the earth given that the radius of the earth

Re = 6380 km. Use $g = 9.81\ m/s^2$ and

$G = 6.67 \times 10^{-11}\ Nm^2/kg^2$

Solution

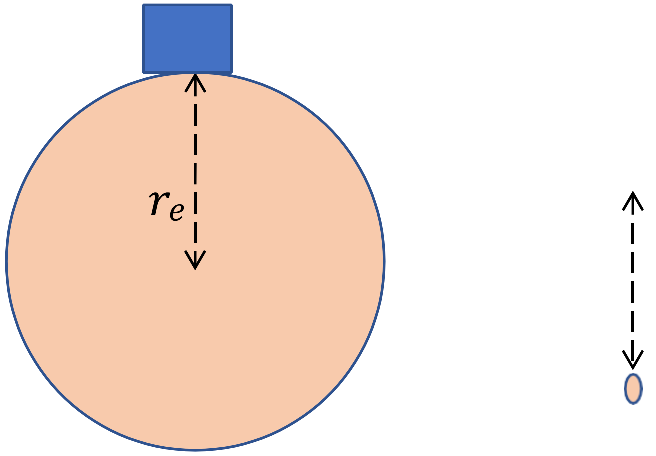
$m_e = \frac{r_e^2 g}{G} = \frac{(6380 \times 10^3)^2 \times 9.81}{6.67 \times 10^{-11}}$

$m_e = 5.96 \times 10^{24} kg$

Acceleration due to gravity relative to the center of earth

- Earth’s density varies with depth
- Is the earth was considered to be made up of spherical shells of equal density, it can be shown that :
- Acceleration due to gravity outside a spherical shell of equal density is the same as it would be the entire mass of the shell were at its center

- The acceleration due to gravity at all points inside a spherical shell of uniform density is zero.
- g outside the earth i.e. $r > r_e$
- The gravitational acceleration at the surface of the earth is r_e



- $g = G\frac{m_e}{r_e^2}$ (i)
- The entire mass of earth is as if concentrated at the center of the earth.

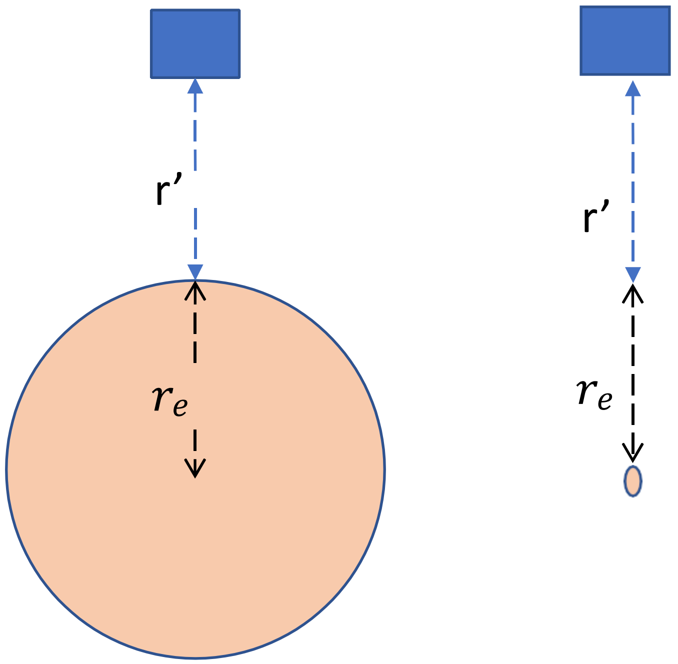
- Consider the body a distance r’ from the earth’s surface.
- The distance of the object from the center of the earth is now $\mathrm{r} = r' + r_e$
- Hence gravitational acceleration g’ be;

m^e

- $g' = G\_2\_$ (ii)

r

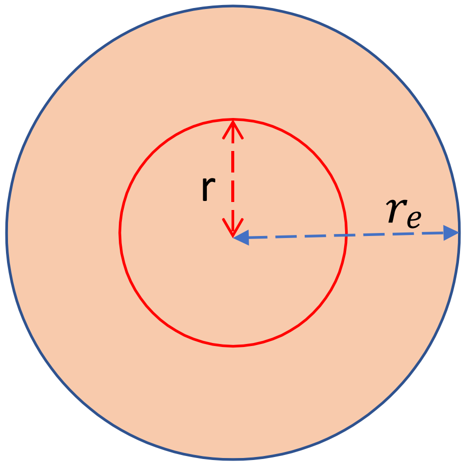
- Dividing eq. (i) and (ii) gives;
- $g' = r\ r\_e2\ 2\quad g$ (iv)
- Hence



- $g' \propto \frac{1}{r^2}$ (v)

- inside the earth $(r < r_e)$

–



- Consider a sphere mass m, radius r inside the earth..
- The gravitational acceleration;
- $g' = G\frac{m}{r^2}$ (vi)
- Now, if earth has a uniform density ρ,

then; • Now, the acceleration due to gravity

- $4 \quad 3\,4 \quad 3$ at the earth's surface is;

$m = \frac{4}{3}\pi r^3 \rho$ and $m_e = \frac{4}{3}\pi r_e^3 \rho$ • $g = G\frac{m_e}{r_e^2}$ (ix)

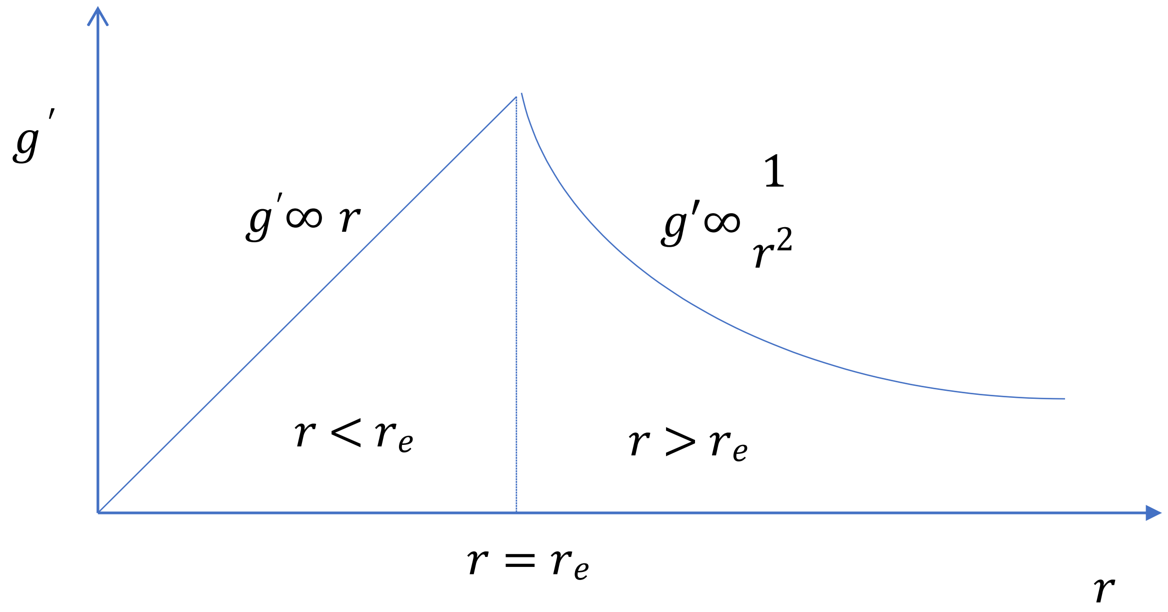
- Dividing the masses
- $\frac{m}{m_e} = \frac{r^3}{r_e^3}$ (vii) • Using eq. (vi) in (v) leads to;

- $g' = {}_{re}\ g$ (x)
- Using$\overline{g}' = G$ (vii ^{m}) ^{r}in $=$ (vi ()G) $_{-}$ (viii) •• Hence$g$
- $g\ {}_{e}\quad me\quad 1'\infty\ r$ (xi)

$3 \qquad\qquad 2$

$re \qquad\qquad re \quad re$

- The variation of g’ as a function of r given by equations (v) and (xi) can be represented graphically as;



Escape velocity

- When an object is thrown upwards, it decelerates uniformly to zero, then starts accelerating uniformly towards the point of projection.
- The body decelerates because it is slowed down by the earth’s gravitational force.

- The height reached depends on the speed at which it was projected upward; the greater the speed, the greater the height.
- If the speed is large enough, the object can reach to such height (inifinity) that it does not come down.
- The minimum speed at which this happens is called escape velocity.
- Now, consider a body mass m being projected upwards with a velocity v form earth.
- When the body is at a distance r from the center of the earth, the gravitational force on the body will be.
- $F = G\frac{m_e m_2}{r}$ _(i)
- The small work dW done in moving the body through a small distance dr against gravitational force is;

- $dW = G\dfrac{m_e m}{r^2}\,dr$ (ii)
- The total work done in moving the object from the earth's surface $(r = r_e)$ to infinity $(r = \infty)$ is equal to:
- $W = \displaystyle\int_{r_e}^{\infty} G\frac{m_e m}{r^2}\,dr = G m_e m\left[-\frac{1}{r}\right]_{r_e}^{\infty}$ (iii)
- $W = G\dfrac{m_e m}{r_e}$ (*)
- If the body is able to do this work and escape then it must have kinetic energy equal to:

- $KE = \frac{1}{2} mv^2$ (iv)
- Equating eq. (iv) and (iii) gives the escape velocity;
- $\frac{1}{2} mv^2 = G\frac{m_e m}{r_e}$ (v)
- $v = \sqrt{\frac{2Gm_e}{r_e}}$ (vi)
- Substituting the known values give $v = 11000\ m/s$

NOTE

- Equation (vi) applies to bodies that are not driven (powered) and only rely on the velocity of projection.

- Escape velocity is independent on the path of projection since KE lost depends on the heigh reached and not on the path.

… END …